

Thinking Outside the Box: Non-Designer Perspectives and Recommendations for Template-Based Graphic Design Tools

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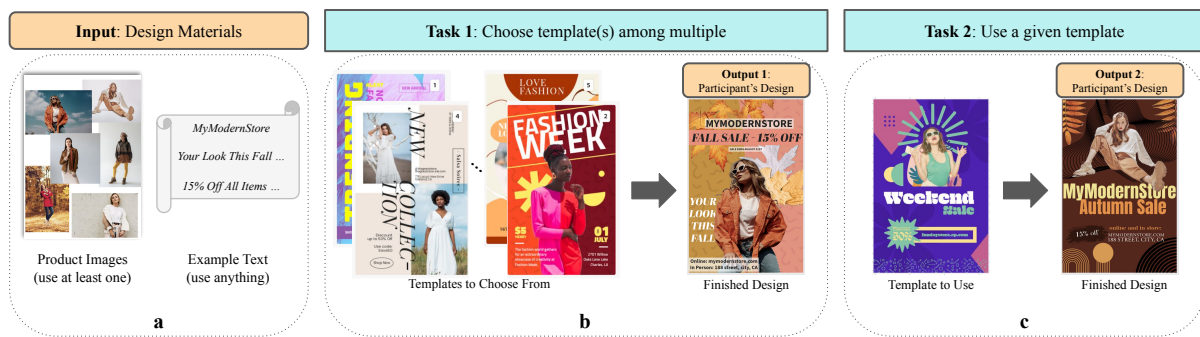


Figure 1: The think-aloud study process flow. In a within-subject study, each participant was asked to create an advertising poster for a fictitious clothing store. a) Product images and some example text were initially given to the participant. b) *Task 1* provided the participant with 9 templates to choose from. c) *Task 2* required them to use a particular template. In both tasks, participants edited the template(s) to create their own design (output 1 and 2, respectively).

ABSTRACT

Many digital graphic design tools provide design templates to help non-designers quickly get started. Users often have assets (e.g. images and text) that they want to incorporate when starting their design. Existing interaction schemes may be insufficient to help users adapt a template to work with their assets and hinder their sense of creativity and productivity. To investigate the effects of templates on non-designer workflows, we conducted a think-aloud design study (N=10) where participants designed posters using one or more templates, followed by semi-structured interviews. Our findings show that while templates help users converge to a final design quickly and partially alleviate the need for tedious decision-making, they may also lead to fixation and frustration when interactions do not support divergent thinking and flexible editing. We present design recommendations for creativity support tools (CSTs) that make use of templates, to address non-designers' creative and practical needs.

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CCS CONCEPTS

• **Human-centered computing** → User studies.

KEYWORDS

Template-Based Graphic Design, Non-Designer Experience, Creativity Support Tools, Qualitative Methods

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1 INTRODUCTION

Providing professional-looking templates is a popular way of scaffolding and guiding non-professional designers in the graphic design process. Templates are often curated and presented in graphic design tools (e.g., Adobe Express, Canva, PowerPoint) to help users get inspired and provide a starting examples for their design process. Research has shown that examples—i.e., materials or previously created artifacts that can contribute to the design process—help professionals in various steps of design, by providing a framework or direction for the project, facilitating ideation and generation

of new ideas, or providing a reference for evaluating generated solutions [10, 11, 24]. Templates have also been shown to be helpful in guiding users through a predefined path to valid solutions [31] by providing initial constraints for the vast design space [28], thus mitigating the paralysis from facing too many possibilities and the 'blank page' effect [14]. In graphic design tools, templates for posters, fliers, etc. are provided as complete, one-page documents with example text and image assets, and users are encouraged to edit them for their own purposes. In this work, we focus on understanding the template-assisted design workflows of non-designers, defined as individuals with no professional background in graphic design, who engage in design for everyday purposes (e.g., creating a birthday card, an academic poster, or a flier for an event).

Previous research has shown that while templates are valuable as design starters [25], they may also cause restrictions in users' creative explorations [11, 24]. In the context of graphic design, one such scenario may occur when users enter the process with their own visual or textual assets and ideas, but face challenges with adapting the existing template into a form that fits these assets. It is unclear whether current interactions with templates are supporting non-designers with these issues. Moreover, there is a lack of insight into how the selection and use of templates in graphic design tools are steering the creative process for these users and affecting their productivity. This work bridges these gaps by investigating non-designers' use of templates, understanding the pros and cons of current interactions with templates in shaping design workflows, and exploring opportunities to improve graphic design tools to meet user needs more effectively. Our research questions include: **RQ1**) How do interactions with templates affect users' design flow, sense of creativity and productivity? **RQ2**) What issues do non-designers experience when integrating their own assets into expert-made templates? and **RQ3**) How might template-based design tools be designed to support these users in their workflow, while addressing current issues?

To address these questions through a case study, we used Adobe Express¹, a graphic design tool that provides templates as a starting point (Appendix A). We recruited 10 non-designers for participation in a within-subject think-aloud study, who completed two poster design tasks using the tool. To understand usual non-designer workflows, we created a scenario in which a user is assigned a poster project and provided with some assets and goals. In one task, users had access to nine templates to choose from, while in a second task, they were required to start from one pre-selected template. This structure helped us examine how access to multiple design instances may inspire users with further ideas, and investigate the role of template selection in their workflows. After each design activity, participants were interviewed about their experience and reflected on their design process.

2 RELATED WORK

Use of Examples in Creative Work. Research in design and analogical reasoning has assessed how the use of examples can affect the design process and outcomes [2, 32]. Expert designers make use of examples for inspiration in multiple stages of design [10]. However, exposure to examples can lead to design fixation, which occurs

when the designer reproduces features of the examples frequently, leading to limited exploration and output generation [13]. Studies have shown that examples that are more similar to the problem at hand and are more familiar to the designer tend to cause more fixation [32]. Additionally, how examples are used and the structure of the iteration process can influence the occurrence of fixation. For instance, being exposed to many examples can encourage people to infer design principles via comparison-based reasoning, which mitigates fixation compared to using only one example [8, 26]. Studies on divergence have also shown the positive effects of parallel prototyping (i.e., creating multiple initial designs prior to iterating) on the diversity of novice designer creations and their self-efficacy by encouraging exploration [7]. Drawing on these insights, we included two tasks in our study as an attempt to understand the effects of exposure to one versus multiple templates on the design process.

Non-Designers and Template-Based Design. Templates, as used in digital creativity support tools (CSTs), are example designs that are similar to users' target solution and ready to use as a customizable creative asset. Previous work has demonstrated the use of templates in scaffolding the design process for novices and non-designers, in domains such as storytelling [15], user experience design [25], online learning [31] and professional writing [11]. Even though these studies demonstrate use cases of templates, they do not explicitly examine potential pitfalls of utilizing them.

Template-based tools aim at jump-starting design creation to reduce the gap between non-designer and expert capabilities, by providing users with an initial design. However, this starting point is often a finished product, leaving them little room for creative exploration. Templates may hence lead to fixation and feelings of inadequacy as users try to fit their content to the design [1, 38]. Yet, it is unclear how user interactions with template-based tools mediate these effects. Some researchers have proposed tools that provide automated suggestions or generation capabilities when creating and editing visual documents [9, 12, 29, 43, 44]. Most commercial tools (e.g., PowerPoint, Adobe Express and Canva) also support intelligent recommendations. However, little is known about how automated suggestions can assist users in the process of editing a template into a new design. To address these gaps, the present work utilizes a think-aloud study to examine the impact of templates on non-designer workflows.

3 METHODS

The goal of our study was to understand how templates impact non-designer workflows, what issues these users face in the process and whether their sense of creativity and productivity is impacted by the use of templates.

Recruitment & Participants. We deployed a convenience sampling strategy and recruited 10 participants (3 male, 7 female; ages 21-32; 8 graduate and 1 undergraduate student in computer sciences, and 1 law student) from among employees of a large technology company through group messaging channels. Our study followed a protocol approved by an internal review team and no compensation was included. None of the participants identified as professional graphic designers nor had they been trained for this purpose, aligning with our definition of non-designers. Most

¹new.express.adobe.com

had minimal experience designing posters using Canva, Microsoft PowerPoint or similar tools, and none had used Adobe Express.

Procedure. Each study session took 80 minutes to complete. The first 10 minutes involved an on-boarding activity (Appendix B). We then presented the study design scenario, which involved two standardized tasks where participants were asked to design advertising posters, as explained below. We instructed participants to think out loud, and conducted semi-structured interviews after finishing each task. The interviews probed participant template and design choices, points of frustration and satisfaction, and perceptions of creativity and productivity (Appendix C).

The poster design scenario was developed to simulate a real-world activity and understand organic behaviors of non-designers with template-based tools. Participants were asked to design an advertising poster—a form of graphic design commonly used to promote a product [19]—for a fictitious clothing store, and advertise its new collection of garments for the upcoming autumn. A set of design requirements were provided for the scenario, and a fixed set of assets (e.g., images, vector graphics and text). Two tasks were then conducted, where *Task 1* provided participants with 9 templates and *Task 2* required using one given template. To maintain consistency, we provided fixed template(s) for all participants in each design task. Details on the procedure can be found in Appendix B. Figure 1 illustrates the design scenario and the two tasks that participants completed, and Figure 4 shows the Express project environment during the study.

Analysis. To analyze qualitative data from interviews and think-aloud activities, we used Rapid Qualitative Analysis (RQA) [36], a description of which can be found in Appendix D. We report the identified themes in section 4.

4 RESULTS

4.1 Designing with Templates: Two Approaches

Participant behaviors when selecting and using templates were analyzed in *Task 1*, where they had the option of using multiple templates. Participants took two general approaches to leveraging templates: **customization approach**, where they use a template as a guide and add edits in place, and **mix-and-match approach**, where they borrow assets and ideas from multiple templates and combine them in a blank canvas or a template (RQ1).

Figure 2 shows these design approaches along with two participant designs for illustration. P6, for example, began the design process by choosing a template that they thought already looked modern and chic, replaced the images and began iterating on changes to color theme, text, and embellishments (Figure 2.a). On the other hand, P4 observed that one template was more suitable for the target audience given its modern, boxy layout, while another matched the fall vibes requirement due to its brown-orange color palette. Hence, they first tried replacing the model image in the brown-colored template with the product images in the scenario, while taking inspiration from the layout of the boxy template, resulting in an intermediate design (Figure 2.b). The participant then felt that the embellishments in the brown-colored template were too curvy for this boxy layout. Therefore, they created a new page, copied some assets from their intermediate design and made manual edits while having a brown-orange color palette in mind, reaching

the final design (Figure 2.b): *"You already have all these inspiration pieces (template options) here, but it's like you want to bring the parts that are particularly meaningful to you in one place"* [P4]. In this case, the collection of two shortlisted templates served as creative inspiration for the participant to create their own design.

4.2 Template Selection as a Design Stage

To choose template(s) among the available ones in *Task 1*, participants spent a few minutes browsing through the available templates, discussing how much they match the project requirements or their personal preferences. When reflecting on their choice of template(s), most emphasized that they tried to minimize the changes they would have to make after choosing a template: *"I consciously went for a template where I wouldn't have to change too many things"* [P1]. For some, this decision-making was based on the color theme of the template being relevant to an autumn palette or the subjective "modern" look of the template. Some participants mentioned that they chose a template that is *"not too busy"* and *"closest to a blank canvas"* [P2], so as to not make many decisions about removing or keeping elements.

In this controlled study, deciding to use a particular template was a one-time step. In a longer real-world design process, a user may determine that their choice of template was not optimal and iterate on that decision along with other edits. This was raised as a concern: *"I had a choice between a lot of templates and didn't know if I chose the right one"* [P2]. One participant felt they made an error in this step, unable to recover due to time restrictions and the complexity of doing so: *"I missed that the audience are teenage girls, and I had already started working on that template, which is better for an older age group"* [P8]. Not being able to change the template selection led to further avoidance from taking risks in their design: *"I worried I might...make it ugly, so I played safe and didn't add embellishments."* [P8]. Furthermore, *Task 2* posed a challenge of editing a template that may not be in line with their preferences to start with: *"I wanted to use more than one product image. For Task 2 I had to invent a strange way of fitting them in the design while the template I chose in Task 1 already had a good layout"* [P1]. These findings indicate that choosing a template is an important stage in the design workflow, such that having no choice in this regard (as in *Task 2*) or choosing the wrong template (in *Task 1*) hindered participants' success and extent of exploration in later stages (RQ1).

4.3 Templates: Facilitating vs. Constraining Exploration

We identified a number of ways in which non-designers may feel constrained in exploring the design space (RQ2).

Lack of Space for Divergent Thinking. Participants with higher motivation to create novel designs by mixing multiple aspects of templates felt constrained by the existing interactions, as they had to move back and forth between pages to view multiple templates and draw inspiration. One participant suggested that they would rather have a curation space within the tool, and see all templates and assets of interest in one frame or move between them seamlessly: *"I want my own curated mood board somewhere"* [P4]. Most

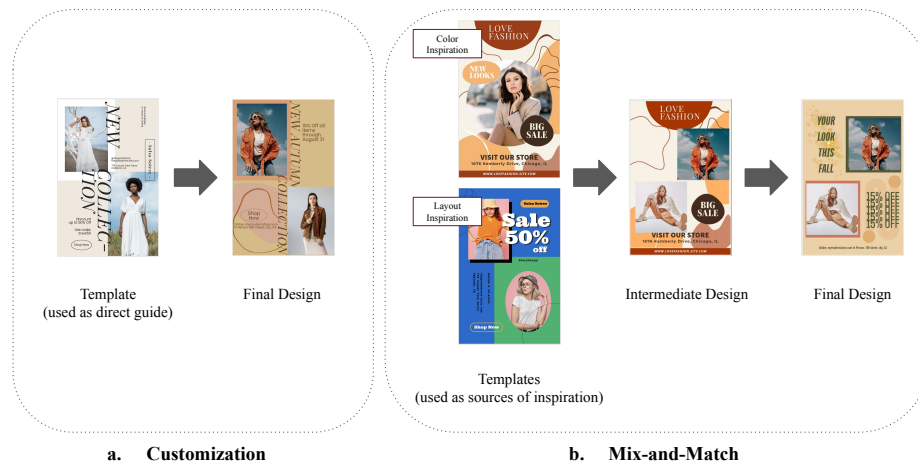


Figure 2: Two main approaches to using templates were observed in our study, illustrated with examples from two participants who: a) used a template as a guide for design principles by choosing one and modifying it for their own purpose (customization), b) used multiple templates for inspiration and brought pieces or ideas together from templates into their own design (mix-and-match).

existing graphic design tools prescribe a more linear approach, assuming that users choose a template first and then make iterative edits. We observe these assumptions are not well-suited to support users who instead leverage templates for inspiration using a mix-and-match approach.

Feeling Restricted by a Complete Design. Templates in both tasks were found to be well-designed around their original assets, making them difficult to adapt to the new assets that participants intended to use. P3 highlighted this by explaining that their chosen template in *Task 1* had a “modern, muted look” that they liked and wanted to keep, and that look was emphasized through the model images’ backgrounds which showed light blue skies and dull nature scenes. Replacing the template images with the required images disrupted that look. To work around this, the participant decided to change the template’s color palette entirely. Many also used the template as a guide for layout, which sometimes came with frustration: “It’s just a template and you have to adapt things to your own project. But I feel it’s harder to do that from a template because you’re trying to make ‘that thing’ work, like shoehorning it, and sometimes it’s just easier if you start from nothing” [P5]. Frequently, participants felt stuck finding a good place for their images or text, and ended up either sticking to the original template layout and “playing safe” [P8] or editing things and not being satisfied as if “something feels off” [P7]. Participants experienced frustration and low confidence from not knowing how to edit a template and feared making a mistake as the template looked too final: “I’m frustrated with myself...I’m just not a good designer” [P8]. To address this, a few expressed the need for a trial and error space outside of the main canvas, such as a drawer for storing candidate assets, or multiple canvases for alternative designs to avoid irreversible edits.

4.4 The Creativity-Productivity Trade-Off

Perceived Productivity. Most participants described templates as a great starting point and time-saver. For some, starting a design

from scratch would be daunting: “I had [the template] to work with and wasn’t starting from a blank canvas, which can be very stressful...It already tells you if you put some image here and some text here, it works” [P5]. This start was deemed valuable especially as participants were designing for practical communication, where they have a goal of what they want to communicate but need help on the design communication. The templates therefore generally led to a sense of productivity.

Participants had varied opinions about what made them feel creative. We report findings by breaking them down into two factors that may have impacted their sense of creativity, based on our findings.

1) Creativity and the Extent of Contributions. Some participants reported that using a template did not lessen their sense of creativity in cases where they tried a number of design ideas and moved or replaced assets [P1, P5]: “Here, I tried a bunch of things and moved things around so I felt creative” [P5]. Conversely, others felt the template made them feel lazy and less creative as it already included most necessary assets and did not need much change [P6, P9]. Thus, participants generally felt more creative when they made more edits to the initial templates. On the other hand, participants who were more interested in mixing design elements from multiple templates (mix-and-match, subsection 4.1) reported feeling most creative when exposed to a variety of templates (i.e. in *Task 1*).

2) Creativity and the Quality of Output. A number of participants considered an aspect of creativity that is related to the quality of their final design [P3, P5, P6, P8]. In particular, they felt more creative when their design “looked more creative” [P8]. Indeed, the fact that the templates were designed by experts helped some participants think about styles that they “otherwise would have not considered” [P6], such as novel ways of overlying assets. P8 emphasized: “...because I built on the template the result looks better than starting from scratch, so I feel more creative in that sense” [P8]. Even though they admitted to not have contributed much to these ideas,

the exposure to novel design paths gave them a sense of creativity and satisfaction with the result. Participants who had this view did not report a difference in creativity between *Tasks 1* and *2*, possibly due to the perception of creativity being oriented toward the end result as opposed to the design process (RQ1).

Importantly, not only most participants felt more productive due to the existence of templates, but they actively made choices that would minimize future efforts. As discussed in subsection 4.2, a majority of decisions with regards to the choice of template(s) and edits were made based on the extent to which they would alleviate the need for edits or major decision-making down the line: *"I felt a bit less creative because I get lazy and don't want to think for myself what I do about the assets. But I don't feel it's a bad thing for me, it just increases my productivity"* [P9]. Hence, perceptions of productivity can be in conflict with the user's extent of contribution, which itself impacts users' sense of creativity. Understanding this trade-off may help with designing CSTs that optimize both of these notions, while taking into account the user's practical needs (e.g., designing in time crunch vs. leisurely).

5 DISCUSSION AND RECOMMENDATIONS

Our study highlighted important issues that non-designers face when using templates. We discuss our findings from an interaction design perspective and recommend ways to improve current template-based graphic design tools (RQ3).

5.1 Support decision-making while recognizing different types of users

While current tools support automated suggestions, they often lack an awareness of the current design or project requirements, and scrolling through all recommended options is tedious. Many non-designers therefore use templates as a guide for these principles by inferring expert design decisions from them. Our findings show that choosing an appropriate template itself is a complex stage of the design process that deeply impacts other steps and decisions. Studies of design fixation have also shown that an early choice of the example design can lead to limited exploration in the later stages of design [13]. Even when chosen, templates are only the start, as users have to adapt them into a more suitable design for their particular project assets. Providing design feedback [42] and dynamically-updated recommendations for the template selection, fonts [9, 33], colors [34], layout [29], and assets [17] are among potential solutions to these problems. Enabling users to filter the provided options in a tool's library may also give them a more precise control, e.g. by adding tags such as 'modern' to fonts that are often used in modern-looking posters.

Interactions with current template-based graphic design tools often assume that users will choose a template, use it as a direct guide or stencil, and customize it for their purposes [20, 38]. Most suggestions and decision support mechanisms (e.g., font suggestions), therefore, come into play after one template has been selected. Inconsistent with this paradigm, we found that users may take multiple approaches to leveraging templates. Users with a mix-and-match approach, for instance, may benefit from suggestions on how to combine favorite assets they have gathered into a novel design. Such ideation aid can be provided by using automated approaches

similar to [18] and [41], through which example designs can be retargeted into similar prototypes, or various features of multiple examples can be combined to create new designs. On the other hand, users with a customization approach may prefer to select one template and choose among various layouts suggested for their design. More extensive studies with a larger number of participants are needed to understand variations of user preferences and types, and how they may best be supported by CSTs.

5.2 Encourage exploration through supporting divergent thinking and iterations

Templates can be used as inspiration, e.g. by providing example assets, fonts and styles that may be applicable to a user's project. In our study, this behavior was observed mostly among participants with a mix-and-match approach, and we found that the *choose template-edit* paradigm felt confining and frustrating to them. As suggested by participants, one approach to facilitating divergent thinking and encouraging exploration of various assets is the use of a mood board for collecting assets. This finding is consistent with previous work that supports the effectiveness of example curation in supporting idea generation [16, 22, 23, 40]. A moodboard space may help users browse through the template options as filtered by their project requirements, collect pieces that are meaningful to them, and use ideas from favorite templates to make decisions about their own design. One such approach can be found in [20], where the proposed tool allows users to begin design from one template while borrowing ideas from recommended examples. Including multiple canvases for alternative user designs is another approach to facilitating divergence and exploration, grounded in studies that demonstrate the benefits of parallel prototyping in diverse idea generation [7].

Design is an iterative process. A user may experiment with a design idea and realize that it does not fit the project, their own preferences, or design aesthetics [3, 4, 35]. Time crunch can also increase the pressure users feel when errors occur. Hence, it is imperative that computer-based tools account for smooth recovery from errors and iterations [30]. In the case of designing with templates, this can be done by allowing multiple iterations of one's own work to be seen and used, which has been shown to aid sense-making and reflecting on design ideas for non-designers [23]. Creating a sandbox space or an asset drawer where users can store materials for later use and adding hide/unhide controls for layers may reduce their fear of losing materials. Finally, developers of CSTs can integrate version control capabilities into these tools (as proposed in [37]) to reassure non-designers that their decisions can easily be reverted, thus encouraging creative risk-taking in the absence of design expertise.

5.3 Balance productivity and creativity by identifying meaningful tasks

In general, we observed that some non-designers experience fixation as a result of using templates and become resistant to changing the original templates due to the completeness of the design, in line with previous research [6, 25]. Non-designers in our study considered two factors that impact creativity, including the quality of their output design (similar to prior work on the associations

between the aesthetic appeal of generated ideas and how creative they are perceived [5]) and their own level of contribution and edits (consistent with studies on the role of engagement in creativity [27]). That said, making more edits often takes longer and compromises productivity. In our study, productivity itself was found to be a benefit of using templates, pointing to a potential trade-off between these two concepts.

To balance users' needs for both productivity and creativity, CSTs can identify meaningful tasks that contribute to the design process, such as choosing a color combination. For these tasks, suggestions that can be vetted by the user are often preferred [38]. On the other hand, more mechanical and redundant tasks can be fully automated. An example of these in our study was replacing the original image in a template with a user's image and applying similar formatting as the original, which is essential when using a template but also tedious. Automatic execution of such activities can reduce the cognitive load on users and increase their productivity while allowing them to focus their attention and time on the more important steps of the design, thus facilitating creative thinking.

6 LIMITATIONS AND FUTURE WORK

Our study had several limitations, including a small number of participants. Future work should consider a larger sample size with a range of prior knowledge in design. We also acknowledge that Adobe Express does not represent the full range of commercial tools for template-assisted design. Even though our research questions were independent of the tool's features, e.g., automated recommendations, a variety of tools should be studied to eliminate potential effects of tool-specific features. Validated quantitative measures can be used in future studies to get a more precise understanding and comparison of user perceptions, such as creativity and productivity (and the effects of having one versus multiple templates). Adding an experiment arm where participants do not utilize any templates may also be helpful in understanding the effects of templates on the design process.

7 CONCLUSION

The use of templates is a common, yet under-researched approach to guiding non-designers in graphic design. This paper contributes to understanding the design process with templates and how the current interactions limit or foster creativity and productivity. We found two user approaches in the use of templates, instances of limiting interactions with template-based tools were observed for both approaches, and trade-offs between perceived creativity and productivity were discussed. We presented implications of these results to inform the design of creativity support tools for graphic design.

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REFERENCES

- [1] Marine Agogu , Akin Kazak i, Benoit Weil, and Mathieu Cassotti. 2011. The impact of examples on creative design: explaining fixation and stimulation effects. In *DS 68-2: Proceedings of the 18th International Conference on Engineering Design*

- (ICED 11), *Impacting Society through Engineering Design*, Vol. 2: *Design Theory and Research Methodology*, Lyngby/Copenhagen, Denmark, 15-19.08. 2011.
- [2] Nathalie Bonnardel. 1999. Creativity in design activities: The role of analogies in a constrained cognitive environment. In *Proceedings of the 3rd conference on Creativity & cognition*. 158–165.
- [3] Lydia B Chilton, Ecenaz Jen Ozmen, Sam H Ross, and Vivian Liu. 2021. VisiFit: Structuring iterative improvement for novice designers. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [4] Lydia B Chilton, Savvas Petridis, and Maneesh Agrawala. 2019. VisiBlends: A flexible workflow for visual blends. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [5] Henri HCM Christiaans. 2002. Creativity as a design criterion. *Communication Research Journal* 14, 1 (2002), 41–54.
- [6] Nigel Cross. 2021. *Engineering design methods: strategies for product design*. John Wiley & Sons.
- [7] Steven P Dow, Alana Glassco, Jonathan Kass, Melissa Schwarz, Daniel L Schwartz, and Scott R Klemmer. 2010. Parallel prototyping leads to better design results, more divergence, and increased self-efficacy. *ACM Transactions on Computer-Human Interaction (TOCHI)* 17, 4 (2010), 1–24.
- [8] Mary L Gick and Keith J Holyoak. 1983. Schema induction and analogical transfer. *Cognitive psychology* 15, 1 (1983), 1–38.
- [9] Shunan Guo, Zhuochen Jin, Fuling Sun, Jingwen Li, Zhaorui Li, Yang Shi, and Nan Cao. 2021. Vinci: an intelligent graphic design system for generating advertising posters. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–17.
- [10] Scarlett R Herring, Chia-Chen Chang, Jesse Krantzler, and Brian P Bailey. 2009. Getting inspired! Understanding how and why examples are used in creative design practice. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 87–96.
- [11] Julie S Hui, Darren Gergle, and Elizabeth M Gerber. 2018. Introassist: A tool to support writing introductory help requests. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [12] Ali Jahanian and Ali Jahanian. 2016. Automatic Design of Self-Published Media: A Case Study of Magazine Covers. *Quantifying Aesthetics of Visual Design Applied to Automatic Design* (2016), 69–99.
- [13] David G Jansson and Steven M Smith. 1991. Design fixation. *Design studies* 12, 1 (1991), 3–11.
- [14] Caneel K Joyce. 2009. *The blank page: Effects of constraint on creativity*. University of California, Berkeley.
- [15] Joy Kim, Mira Dontcheva, Wilmo Li, Michael S Bernstein, and Daniela Steinsapir. 2015. Motif: Supporting novice creativity through expert patterns. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 1211–1220.
- [16] Janin Koch, Andr s Lucero, Lena Hegemann, and Antti Oulasvirta. 2019. May AI? Design ideation with cooperative contextual bandits. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [17] Balazs Kovacs, Peter O'Donovan, Kavita Bala, and Aaron Hertzmann. 2018. Context-aware asset search for graphic design. *IEEE transactions on visualization and computer graphics* 25, 7 (2018), 2419–2429.
- [18] Ranjitha Kumar, Jerry O Talton, Salman Ahmad, and Scott R Klemmer. 2011. Bricolage: example-based retargeting for web design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2197–2206.
- [19] Robin Landa. 2018. *Graphic design solutions*. Cengage Learning.
- [20] Brian Lee, Savil Srivastava, Ranjitha Kumar, Ronen Braffman, and Scott R Klemmer. 2010. Designing with interactive example galleries. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2257–2266.
- [21] Allison A Lewinski, Matthew J Crowley, Christopher Miller, Hayden B Bosworth, George L Jackson, Karen Steinhauer, Courtney White-Clark, Felicia McCant, and Leah L Zullig. 2021. Applied rapid qualitative analysis to develop a contextually appropriate intervention and increase the likelihood of uptake. *Medical care* 59, 6 Suppl 3 (2021), S242.
- [22] Andr s Lucero. 2012. Framing, aligning, paradoxing, abstracting, and directing: how design mood boards work. In *Proceedings of the designing interactive systems conference*. 438–447.
- [23] Nic Lupfer, Andruid Kerne, Rhema Linder, Hannah Fowler, Vijay Rajanna, Matthew Carrasco, and Alyssa Valdez. 2019. Multiscale Design Curation: Supporting Computer Science Students' Iterative and Reflective Creative Processes. In *Proceedings of the 2019 Conference on Creativity and Cognition*. 233–245.
- [24] Stephen MacNeil, Zijian Ding, Kexin Quan, Thomas j Parashos, Yajie Sun, and Steven P Dow. 2021. Framing Creative Work: Helping Novices Frame Better Problems through Interactive Scaffolding. In *Creativity and Cognition*. 1–10.
- [25] Stephen MacNeil, Ziheng Huang, Kenneth Chen, Zijian Ding, Alexander Yu, Kendall Nakai, and Steven P Dow. 2023. Freeform Templates: Combining Freeform Curation with Structured Templates. In *Proceedings of the 15th Conference on Creativity and Cognition*. 478–488.
- [26] Richard L Marsh, Joshua D Landau, and Jason L Hicks. 1996. How examples may (and may not) constrain creativity. *Memory & cognition* 24, 5 (1996), 669–680.

- [27] Marian Garcia Martinez. 2015. Solver engagement in knowledge sharing in crowdsourcing communities: Exploring the link to creativity. *Research Policy* 44, 8 (2015), 1419–1430.
- [28] C Page Moreau and Darren W Dahl. 2005. Designing the solution: The impact of constraints on consumers' creativity. *Journal of Consumer research* 32, 1 (2005), 13–22.
- [29] Peter O'Donovan, Aseem Agarwala, and Aaron Hertzmann. 2015. Designscape: Design with interactive layout suggestions. In *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 1221–1224.
- [30] Srishti Palani, David Ledo, George Fitzmaurice, and Fraser Anderson. 2022. "I don't want to feel like I'm working in a 1960s factory": The Practitioner Perspective on Creativity Support Tool Adoption. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [31] Vineet Pandey, Justine Debelius, Embriette R Hyde, Tomasz Kosciolk, Rob Knight, and Scott Klemmer. 2018. Docent: transforming personal intuitions to scientific hypotheses through content learning and process training. In *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*. 1–10.
- [32] A Terry Purcell and John S Gero. 1991. The effects of examples on the results of a design activity. In *Artificial Intelligence in Design '91*. Elsevier, 525–542.
- [33] Jocelyn J Shen, Kathryn Jin, Ann Zhang, Cynthia Breazeal, and Hae Won Park. 2023. Affective Typography: The Effect of AI-Driven Font Design on Empathetic Story Reading. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [34] Xinyu Shi, Ziqi Zhou, Jing Wen Zhang, Ali Neshati, Anjul Kumar Tyagi, Ryan Rossi, Shunan Guo, Fan Du, and Jian Zhao. 2023. De-Stijl: Facilitating Graphics Design with Interactive 2D Color Palette Recommendation. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [35] Sungbok Shin, Sanghyun Hong, and Niklas Elmquist. 2023. Perceptual Pat: A Virtual Human Visual System for Iterative Visualization Design. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [36] Sara M St. George, Audrey R Harkness, Carlos E Rodriguez-Diaz, Elliott R Weinstein, Vanina Pavia, and Alison B Hamilton. 2023. Applying rapid qualitative analysis for health equity: lessons learned using "EARS" with Latino communities. *International journal of qualitative methods* 22 (2023), 16094069231164938.
- [37] Sarah Stermann, Molly Jane Nicholas, and Eric Paulos. 2022. Towards Creative Version Control. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–25.
- [38] Kesler Tanner. 2019. *Visual design tools in support of novice creativity*. Stanford University.
- [39] Beck Taylor, Catherine Henshall, Sara Kenyon, Ian Litchfield, and Sheila Greenfield. 2018. Can rapid approaches to qualitative analysis deliver timely, valid findings to clinical leaders? A mixed methods study comparing rapid and thematic analysis. *BMJ open* 8, 10 (2018), e019993.
- [40] Qian Wan and Zhicong Lu. 2023. Investigating Semantically-enhanced Exploration of GAN Latent Space via a Digital Mood Board. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–5.
- [41] Jeremy Warner, Kyu Won Kim, and Bjoern Hartmann. 2023. Interactive Flexible Style Transfer for Vector Graphics. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–14.
- [42] Anbang Xu, Shih-Wen Huang, and Brian Bailey. 2014. Voyant: generating structured feedback on visual designs using a crowd of non-experts. In *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing*. 1433–1444.
- [43] Xuyong Yang, Tao Mei, Ying-Qing Xu, Yong Rui, and Shipeng Li. 2016. Automatic generation of visual-textual presentation layout. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 12, 2 (2016), 1–22.
- [44] Shuyu Zheng, Ziniu Hu, and Yun Ma. 2019. Faceoff: Assisting the manifestation design of web graphical user interface. In *Proceedings of the Twelfth ACM International Conference on Web Search and Data Mining*. 774–777.

8 APPENDICES

A OVERVIEW OF ADOBE EXPRESS

Adobe Express has semantic search and browsing functionalities to find templates that relate to any topic (e.g. wedding invitation) for any type of one-page document (e.g. poster, flier, social media post, etc.), along with a stock of searchable assets that may be used to customize the template. A user may, however, decide to begin a design from scratch, which can be done by creating a blank page and adding one's assets to it. For each image, icon or graphic asset that a user includes, multiple formatting options exist, including background removal, recoloring, cropping into various shapes, etc. Text content can be formatted using font selection, sizes and artistic

effects, and a list of recommended fonts are provided to the user when they select a textbox. At any point during the design process, a user can also change the background color of their page, or select a color theme among multiple options with semantic tags (e.g. retro or classic) which recolors all shapes, background and text with a palette that is associated with that tag. Users can also move, resize or rotate all assets, or transfer them to various layers of the design. A Screenshot of this tool can be found in Figure 3

B STUDY PROCEDURE DETAILS

At the beginning of each session, an on-boarding activity was performed which involved using a poster template for charity donations, and editing its text, images and colors with basic manual editing features and intelligent recommendations. The researcher first explained functionalities of Adobe Express and provided a practice design task. The participant was then given a chance to perform this task and interact with the functionalities. Once the participant had exposure to important features of Adobe Express, we conducted the standardized tasks using a poster design scenario.

Participants were asked to design an advertising poster for a fictitious clothing store, and advertise this store's new collection of garments for the upcoming autumn. Some assets (i.e., images and text) were provided, as well as a set of design requirements. The requirements for the design tasks included: targeting customers who are in the age range of 15-21, have a modern and chic and warm look and feel, convey autumn vibes, and a confident tone of text.

An advertising scenario for clothing was chosen for our study because: 1) it requires minimal artistic or engineering ability, 2) it provides creative constraints for participants by presenting key communicative requirements, thereby reducing the design space to some extent, and 3) there is nonetheless a degree of aesthetic exploration needed for designing advertising posters [9], which allows our results to expose various pain points and design approaches in one session.

The selection of assets provided to the participants included: 1) six product images (models wearing the garments) from which at least one was required to be included in the design, 2) some example text that can convey the message and holds the website and address of the store (text included seven lines from which participants could choose, but they were also allowed to come up with their own text, so long as the goal of the poster was met), and 3) 50 optional assets with photographic or graphic elements which could be used for embellishment. The materials provided to the participants are shown in Figure 1, except for the optional embellishment assets due to license restrictions.

B.1 Tasks

Users completed two design tasks that involved templates, each in a maximum of 30 minutes. The order of tasks was counterbalanced to avoid bias from carry-over effects.

Figure 1 shows the tasks along with the templates and assets that were used in each. The templates did not meet all the requirements of the project to encourage participants to edit the design as needed. Participants were not restricted on how they should use the template(s) to complete each task (e.g., edit a template in place or

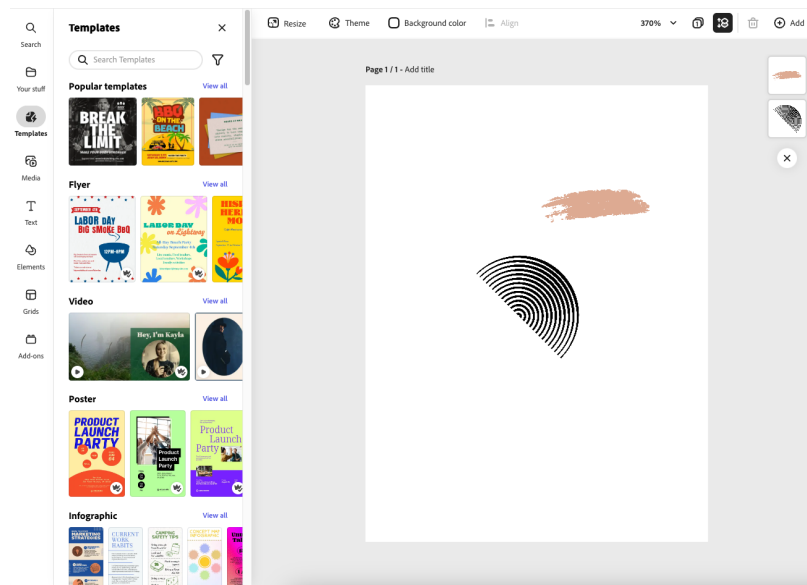
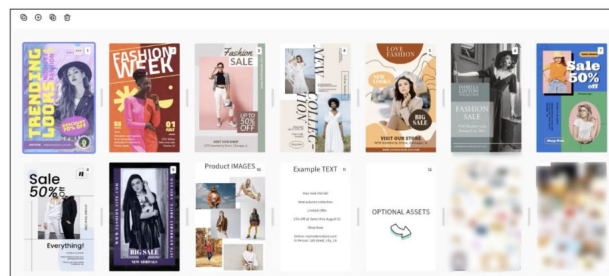


Figure 3: Adobe Express Environment. The left pane shows a stock of templates that a user can choose from and customize for their own purpose. The design canvas is in the center, where the user may add assets to. Layers of assets can be set from the far-right pane, so that one asset is on top of another.

a) Project View



b) Canvas

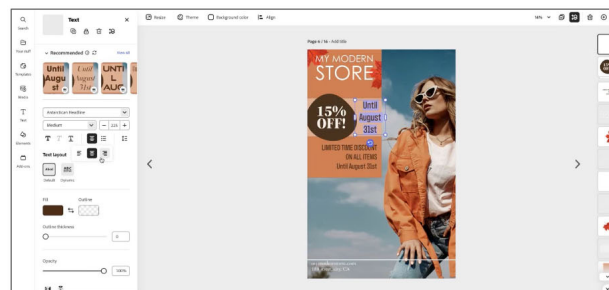


Figure 4: Participant project environment in Adobe Express. a) multiple canvases, including the curated template and asset collections in the project view (optional assets are blurred due to license restrictions), b) canvas for participant design

copy parts of it in a new page). We recorded the time taken to finish each task. Paired-sample t-tests on this time showed no significant difference between *Task 1* and *Task 2* across all participants.

Task 1: This task involved choosing among a set of 9 pre-selected templates the user could use to start their design.

Task 2: For this task, participants were given one pre-selected template to use as their starting point in the design process. They were not allowed to use any other templates for inspiration.

C INTERVIEW GUIDE

- What were your impressions of the design process you just went through?
- Which part of the activity did you enjoy the most/least? why?
- Did you face any challenges in finding a good template?
- How did you decide to choose this template?
- How did you decide which images to include? Was that easy for you or were there times you wish you could get help? How about the text?
- How was the experience of putting the assets on the template?
- How did you know when to replace/edit/move the assets? Did you ever feel stuck while doing that? What could have been helpful to you in this process?
- Did you feel that having the template helped you? Why or why not? Do you think it made you more or less creative/productive?
- If you started over/if you were allowed to, would you have changed your template?

D RAPID QUALITATIVE ANALYSIS

We used Rapid Qualitative Analysis (RQA) as the method for analyzing qualitative data from interviews and think-aloud activities. This analysis method is an applied technique used to gather and analyze actionable, targeted qualitative data within a shorter timeline than traditional qualitative methods [36]. RQA has previously been used in developing and improving health interventions and has shown to be helpful in increasing the relevancy and efficiency of data collection and analysis [21, 39].

For each participant, the first author of this paper created a summary page outlining various domains of interest related to the research questions, such as points of frustration and satisfaction when designing the poster. This page includes important quotes or codes from semi-structured interviews, which gives the researcher insights into potential pain points and ways of addressing them in practice. The content of summary pages for all participants were then populated in an Excel sheet, and matrix analysis was performed to find consistencies and develop recommendations for improving graphic design tools as related to templates.